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FDR-MSA: Enhancing multimodal sentiment analysis through feature disentanglement and reconstruction

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ABSTRACT

Multimodal sentiment analysis (MSA) is crucial as it integrates textual, visual, and audio information from videos to accurately identify human emotional states. This study proposes an innovative multimodal feature decoupling strategy that categorizes sentiment features into common and private features. The private features aim to accurately capture the uniqueness of each modality, thereby increasing feature diversity. In contrast, the common features seek to identify and capture commonalities among different modalities, thus reducing potential information loss during decoupling. To achieve this, we designed dedicated encoders and loss function constraints for both types of features. Additionally, to mitigate information redundancy and prevent key information loss during decoupled representation learning, we introduce a dual feature reconstruction mechanism comprising unimodal feature reconstruction (UFR) and multimodal feature reconstruction (MFR). These mechanisms preserve vital information from the decoupling process and mitigate the impact of redundant data. Our extensive experiments on three datasets demonstrate that our method achieves a significant margin of approximately 1%–3% in accuracy, illustrating that our approach outperforms existing advanced techniques significantly, resulting in noteworthy performance enhancements.

1. Introduction

The rapid advancement of intelligent human–computer interaction technologies has significantly expanded the application scope of multimodal data, driving the progress of multimodal sentiment analysis (MSA) [1–3]. MSA focuses on parsing and understanding human emotions expressed through various sensory channels like voices and videos, encompassing text, visual, and audio signals [4,5]. Integrating multimodal information empowers researchers to comprehensively grasp human emotional states, resulting in more accurate and profound emotional resonance in human–computer interactions. Through extensive exploration and processing of multimodal data, researchers have captured emotional cues from text, videos, and audio, including facial expressions, tone shifts, and body language [6–8]. They have integrated data processing with machine learning technologies to discern human emotional orientations. MSA research, grounded in text, audio, and video modalities, centres around two primary tasks: exploring intermodal interactions to uncover deeper semantic meanings and devising innovative data fusion techniques to create efficient and expressive data representations. These representations significantly enhance the accuracy and efficiency of sentiment analysis.

In the MSA domain, traditional processing methods frequently treat information from diverse modalities in a uniform manner [9,10]. This practice often disregards inherent differences and uneven distributions among modalities, resulting in coarse and redundant feature representations. To tackle this challenge, researchers have focused on extracting desired characteristics from each modality, striving to more precisely capture multimodal data features and optimize the information fusion process. Generally, in the representation learning phase of an MSA scenario, representations are broken down into modality-invariant (common) and modality-specific (private) representations [11,12]. The former encompasses shared attributes among multimodal features, while the latter captures the distinctive traits of each modality. However, the "divide and conquer" strategy encounters challenges in managing the heterogeneity of multimodal data. Variations in features across different data channels, differences in sampling rates, and the uniqueness of each modality all contribute to the complexity of decoupling and integrating effective multimodal representations.

Therefore, we further refine the fine-grained decoupling method for multimodal features, enabling it to capture private features that highlight differences among independent modalities while laying the groundwork for commonality learning through interconnectedness, shown in Fig. 1. For instance, text information is closely associated with lip movement trajectories but does not demonstrate this extent of association with eye muscle movements. Thus, forcibly integrating

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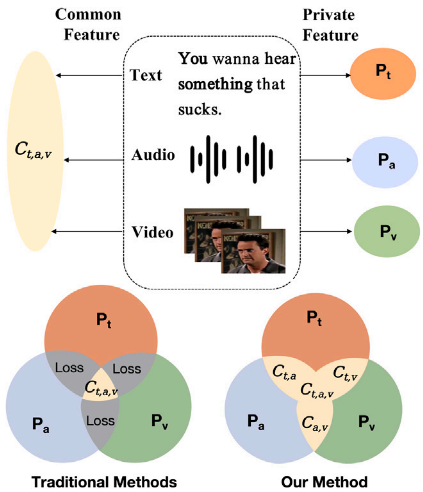

Fig. 1. Traditional methods often overlook common features across modalities. Our approach retains these essential features, capturing crucial information that traditional methods ignore or discard.

irrelevant modalities may yield meaningless information. Commonality learning aims to facilitate information exchange between different modalities by effectively transforming their shared features, which reflects their inherent connections. By employing this decoupling strategy, we achieve feature representations rich in desirable attributes and enhance the quality of multimodal representations. Additionally, we recognize the significant challenge of identifying and removing irrelevant or disruptive information during representation learning, such as facial movement trajectories unrelated to expression analysis. In response, we developed a reconstructive learning method following feature decoupling to eliminate irrelevant information while preserving essential details. Employing a dual representation learning strategy, we decoupled and reconstructed multimodal features, effectively tackling modality-specific and commonality issues stemming from heterogeneous data.

In summary, we propose an MSA framework centred on feature disentanglement and reconstruction (FDR-MSA). This framework effectively mitigates the risks of redundant interferences and information losses while significantly enhancing the accuracy and efficiency of multimodal learning. The developed multimodal learning framework not only improves MSA performance but also opens avenues for exploring novel research directions in multimodal representation learning, demonstrating its broad application potential and opportunities for future development. Our primary contributions can be outlined as follows:

(1) We present an effective method for disentangling multimodal data features, adept at discerning and handling both common and private attributes. Private features target the distinctive traits of each modality, while common features capture shared characteristics across modalities. This strategy enables us to harness the inherent coherence and variability of multimodal data more efficiently.

(2) We present a dual reconstruction learning mechanism for both unimodal and multimodal features. This mechanism enhances the model's capacity to preserve vital information after decoupling and integrating interaction features, thereby enhancing the accuracy and efficiency of feature representation learning.

(3) The proposed FDR-MSA framework undergoes thorough experimental validation on three representative MSA datasets, our method achieves a significant margin of approximately 1%–3% in accuracy, demonstrating exceptional performance.

The remainder of the paper is structured as follows: Section 2 discusses related work and provides a literature review. Section 3 defines the problem and outlines our approach. Section 4 details the experimental design, while Section 4.5 presents a comprehensive analysis of the results. Finally, Section 5 concludes the paper, emphasizing our main findings and contributions.

2. Related works

2.1. Multimodal sentiment analysis

Multimodal sentiment analysis (MSA) has witnessed rapid development within the realm of human–computer interaction research, driven by various algorithmic advancements and research focal points. This field aims to improve machines' ability to comprehensively understand human emotions by integrating information from diverse modalities, departing from traditional natural language processing methods. Early efforts in this domain by Zadeh et al. [6] focused on novel fusion techniques employing Cartesian product operations to merge features from three modalities. However, this pioneering approach introduced high spatial and temporal complexities. Subsequent research by Liu et al. [13] introduced a low-rank tensor fusion network (TFN), prioritizing enhanced fusion efficiency by reducing parameters through low-rank tensors, thus significantly improving data processing efficiency. Technological advancements have shifted research focus towards handling nonlinguistic information in multimodal data. For instance, Rahman et al. [14] integrated a multimodal adaptation gate to adapt bidirectional encoder representations from transformers (BERT) [15] and XLNet [16] models, delving deeper into the applications of pretrained language models in multimodal emotion analysis. The development of bidirectional attention-based fusion network (BAFN) [17] and Self-MM [18] methods signifies a deeper understanding of dynamic relationships between modalities and subtask balance. BAFN captures emotional context within modalities using multimodal dynamic enhancement blocks and text-dominated multimodal dynamic routing mechanisms. In contrast, Self-MM focuses on balancing learning progress among subtasks and refining the handling of modalities.

Recent studies, such as ICDN [19] and Hgraph-CL [20], delve deeper into multimodal interactions. ICDN delineates modal interactions through mapping and generalization, introducing a cross-modal transformer to address time series alignment and missing modality information. On the other hand, Hgraph-CL constructs multimodal and unimodal graphs using contrastive learning to enhance model performance. Hao et al. [21] devised a model for determining the latent distribution of target-opinion pairs by integrating conditional variational autoencoders with a transformer-based classification model. These investigations illustrate the progression of multimodal emotion recognition, evolving from early data fusion explorations to efficient, fine-grained processing methodologies. Each advancement signifies a pursuit of deeper comprehension and improved management of multimodal data.

2.2. Disentanglement learning

Early multiview representation learning methods, such as multikernel methods, aimed to maintain the independence of each view, while others relied solely on a common latent space. However, these approaches were primarily tailored to practical applications. Consequently, researchers introduced the concept of decomposing information into specific latent spaces to simulate common (modality-invariant) and private (modality-specific) data. This allowed for the intricate handling of the complexity of multiview data and adaptation to different data feature types. Salzmann et al. [22] decomposed the latent space into common and private spaces, thereby reducing redundant latent representations. Song et al. [23] utilized multiview latent variable-based discriminative models, employing multichain deconstructed latent conditional models (LvMs9) to explicitly learn common and specific view substructures. Wang et al. [24] extended the latent variable-based linear CCA model to a nonlinear observation model parameterized using deep neural networks, enabling the extraction of private variables and the unsupervised distinction of private representations across multiple views. Hazarika et al. [11] pioneered the integration of multiview fusion methods into MSA, projecting each modality feature into two subspaces. The first subspace is modality-invariant, utilized for learning commonalities across modalities and reducing intermodal differences. The second subspace captures the unique characteristics of each modality, facilitating the comprehensive fusion of multimodal emotional representations. Yang et al. [12] advocated the use of a modality-invariant subspace to explore commonalities between different modalities and significantly narrow distribution gaps between them. In contrast, a modality-specific subspace aims to enhance diversity and capture the unique features of each modality. A modality discriminator was introduced to provide adversarial guidance in common and private parameter learning settings.

Hao et al. [25] devised a joint dual learning system and proposed decoupling feature representations into two tasks: fine-grained aspect-oriented opinion variables and content variables. This approach effectively combines sentiment classification and text generation to optimize the feature interaction process. Li et al. [26] utilized a graph distillation unit (GD-unit) to distil each decoupled part and transfer cross-modal knowledge, acquiring more meaningful features that enhance downstream tasks. Additionally, Li et al. [27] introduced a framework that segregates features into modality and utterance spaces, incorporating a dual-level disentanglement mechanism (DDM), a contribution-aware fusion mechanism (CFM), and a context refusal mechanism (CRM), effectively integrating multimodal and contextual features. However, current multimodal information disentanglement methods face a crucial challenge: ensuring robustness to disentangled features. Although these methods can separate feature spaces, they may lead to the loss of original information and require assistance to recover core information from intermodal interactions. To address this challenge, we propose an innovative multimodal information decoupling and reconstruction method.

3. Methods

This section outlines the proposed FDR-MSA method. The overall architecture of the model and its entire pipeline are illustrated in Fig. 2. We utilize BERT and a gated recurrent unit (GRU) [28] to extract features from multimodal data, encompassing text (X_t), video (X_v), and audio (X_a) data. The corresponding features are $\mathbf{H}_t \in \mathbb{R}^{L_t \times W_t}$, $\mathbf{H}_a \in \mathbb{R}^{L_a \times W_a}$, and $\mathbf{H}_v \in \mathbb{R}^{L_v \times W_v}$, where $L(\cdot)$ indicates the feature length and $W(\cdot)$ denotes the feature dimensionality.

3.1. Multimodal representation

Building on prior work, we integrated the BERT pretrained language model, trained on a large-scale corpus, for feature extraction from text data. To capture the temporal characteristics of audio and video data, we employed a GRU. The GRU enables us to capture crucial temporal information that evolves over time, playing a pivotal role in understanding dynamic and fluid sentiment shifts:

$$H_t = \mathrm{BERT}(X_t) \tag{1}$$

$$H_a = \mathrm{GRU}(X_a) \tag{2}$$

$$H_v = \mathrm{GRU}(X_v) \tag{3}$$

When employing BERT for text feature extraction, we selected $[CLS]$ as the H_t vector, which encapsulates the overall meaning of the input text sentence. Regarding the audio and video modalities, H_a and H_v, respectively, were obtained from the features of the last time step of the GRU. To align these features, we projected them onto the same dimension using a linear layer before proceeding to subsequent steps.

3.2. Feature disentanglement learning

Despite the extensive use of feature extraction encoders in processing multimodal data, they often face challenges related to feature redundancy and interference, especially when dealing with heterogeneous features distributed across various modalities. To address this issue, we refined the feature disentanglement process for multimodal data, emphasizing the distinctions and consistencies among features. Our goal is to enable feature encoders to effectively separate and extract common features across different modalities along with their corresponding private features through learning rules. This approach significantly enhances the capability of encoders to handle diverse data types and capture the similarities and differences between modalities accurately. We decomposed three unimodal features into common and private subspaces, effectively capturing the consistency and specificity of these features. In particular, we employed an encoder comprising two-layer perceptrons with Gaussian error linear unit (GeLU) activation [29] to decouple the features as follows:

$$C_{i\epsilon(t,a,v)} = \Phi(H_{i\epsilon(t,a,v)}, \Theta_{\Phi}) \tag{4}$$

$$P_{i\epsilon(t,a,v)} = \Psi(H_{i\epsilon(t,a,v)}, \Theta_{\Psi}) \tag{5}$$

The common encoder is denoted as $\Phi_{i\epsilon(t,a,v)}$, while the private encoder for individual modalities is represented by $\Psi_{i\epsilon(t,a,v)}$. The parameter set Θ_{Φ} corresponds to common features across different modalities, whereas Θ_{Ψ} is dedicated to modality-specific, exclusive features.

3.2.1. Common features

We introduced a set of encoders designed to extract common features from diverse modalities. Additionally, we employed a cross-modal common encoder and decoder denoted as C_{xy} to capture the commonalities among different modalities based on these shared features—an essential aspect that has previously been overlooked. Each cross-modal encoder and decoder was constructed using a two-layer perceptron with the GeLU activation function. These decoders map features to a common feature space denoted as C_{xy}, where x and y represent the source and target modalities, respectively, and $x, y \in \{t, a, v\}$. Subsequently, each common feature C_{xy} is remapped to its respective modality through its corresponding decoder Y. Furthermore, we utilized a cross-modal common decoder to reconstruct the three modalities, facilitating cross-modal sharing.

$$C_{x,y} = Y_y(H_x) \tag{6}$$

This design enhances the model's capacity to transfer information from the source modality to the target modality, thereby improving its ability to process cross-modal information. For instance, the model can explore correspondences between audio signals and specific textual content or analyse associations between video signals and explicit audio content. When extracting these common features, our goal is to capture features that best represent commonalities across various modalities. The common features extracted from different modalities should exhibit high similarity. Therefore, the primary function of these decoders is to further optimize the information transformation process and fuse different modalities.

$$C = \sum_{i\in\{t,a,v\}} C_i \tag{7}$$

We adopt the smooth L1 loss [30] to evaluate the achieved learning quality:

$$L^x_{com} = \mathrm{smooth}_{L1}\Big(\sum_{m\in\{t,a,v\}} (C_x, C_{x,m})\Big) \tag{8}$$

$$L_{com} = \sum_{m\in\{t,a,v\}} L^m_{com} \tag{9}$$

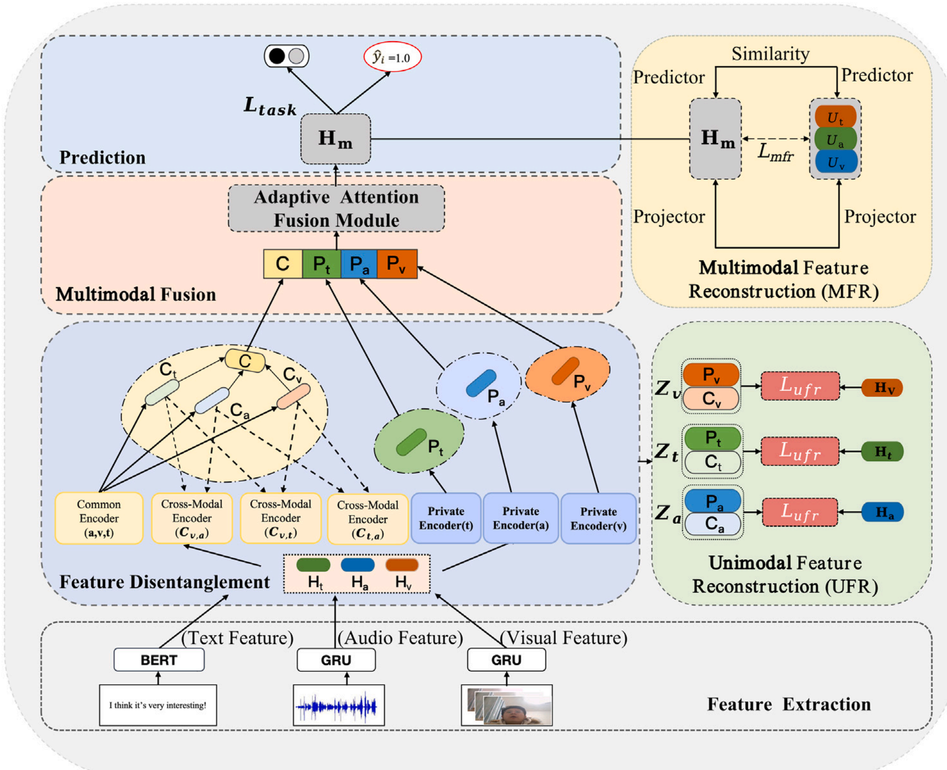

Fig. 2. The overall architecture of the FDR-MSA approach.

3.2.2. Private features

To validate the accurate depiction of specificity in the private features of different modalities, we utilized the Hilbert–Schmidt independence criterion (HSIC) [31] to quantitatively assess the independence among these private representations. Highly independent private features exhibit notable differences, with higher HSIC values indicating stronger specificity. As a result, the model can effectively distinguish and preserve the unique characteristics of each modality during multimodal data processing.

$$\mathrm{HSIC}(P_x, P_y) = (n-1)^{-2}\mathrm{Tr}(\upsilon K_x \upsilon K_y) \tag{10}$$

$$L_{pri} = \sum_{(x,y)} HSIC(P_x, P_y) \tag{11}$$

where $\upsilon = I - \frac{1}{n}\mathbf{e}$, with I being an identity matrix and $\mathbf{e}^T$ representing a column vector of ones. Furthermore, we used the inner product kernel function [32] to compute K_x and K_y, where $x, y \in \{t, a, \upsilon\}$.

3.3. Feature fusion

We employed an adaptive attention mechanism that dynamically adjusts its attention weights based on the importance of each feature:

$$\mu_j = q^T \cdot \tanh(W_j \cdot Y_j + b_j) \tag{12}$$

where q^T represents one shared attention vector, $Y_j \in \{C, P_t, P_a, P_\upsilon\}$, $W_j \in \mathbb{R}^{d_k \times d}$ and $b_j \in \mathbb{R}^{d_k \times 1}$, and the attention values ϕ_j are as follows:

$$\phi_j = \frac{\exp(\mu_j)}{\sum_{j\in\{c,t,a,\upsilon\}} \exp(\mu_j)} \tag{13}$$

A higher ϕ_j indicates greater importance of the corresponding representation. The final fusion representation $F_m \in \mathbb{R}_k^d$ was derived through weighted summation as follows:

$$F_m = \sum_{j\in\{c,t,a,\upsilon\}} \phi_j \odot Y_j \tag{14}$$

Eventually, F_m was passed through the fully connected layers to perform downstream tasks.

3.4. Feature reconstruction

To optimize the effectiveness of the learned features and minimize potential information loss during feature disentanglement learning, we introduce a dual reconstruction strategy. This strategy aims to enhance the efficiency of the feature learning process within the model, consisting of two key components: the reconstruction of multimodal fusion features (multimodal feature reconstruction, MFR) and the reconstruction of local unimodal decoupled features (unimodal feature reconstruction, UFR). By employing these two reconstructive approaches, we could significantly enhance the model's ability to extract crucial information while ensuring the meaningful integrity of the features.

3.4.1. Unimodal feature reconstruction

We employed a simplified decoder structured similarly to the encoder for feature reconstruction. Considering that directly restoring data to the original text, video, and audio would excessively consume the model's capacity and training resources, we selected features H_t, H_a, and H_υ as the reconstruction targets. This approach not only reduces the model's dependency on the original data but also enhances its reconstruction efficiency. To assess the quality of the UFR process, we utilized the smooth L1 loss function:

$$Z_{i\epsilon(t,a,\upsilon)} = [C_i, P_i] \tag{15}$$

$$L_t^{recon} = \mathrm{smooth}_{L1}(H_t - \gamma(Z_t)) \tag{16}$$

$$L_a^{recon} = \mathrm{smooth}_{L1}(H_a - \gamma(Z_a)) \tag{17}$$

$$L_\upsilon^{recon} = \mathrm{smooth}_{L1}(H_\upsilon - \gamma(Z_\upsilon)) \tag{18}$$

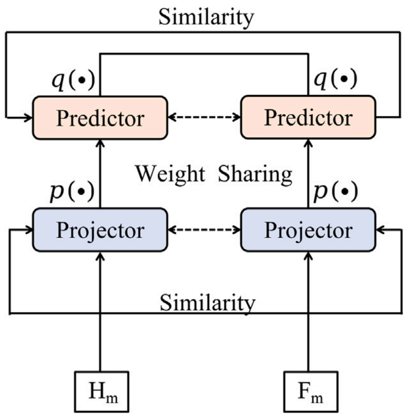

Fig. 3. Illustration of the multimodal feature reconstruction via the SimSiam loss.

where:

$$\mathrm{smooth}_{L1} = \begin{cases} 0.5x^2 & \text{if } x < 1 \\ |x| - 0.5 & \text{otherwise} \end{cases} \quad (19)$$

$$L_{\mathrm{ufr}} = L_v^{recon} + L_a^{recon} + L_t^{recon} \quad (20)$$

Incorporating the unimodal reconstruction learning strategy serves a dual purpose: it safeguards against losing crucial information during the UFR procedure and successfully mitigates the risk of overfitting, thereby enhancing the stability and robustness of the reconstruction process. Reconstructing the features derived from disentangled learning back into unimodal features notably improves the generalization capabilities of the model. Furthermore, this approach demonstrates that the learned features can comprehensively capture and summarize the essential information of each modality, effectively preventing information loss and distortion, thereby significantly boosting the model's overall analytical and predictive capacities.

3.4.2. Multimodal feature reconstruction

Given the inherent complexity of multimodal features, despite the enrichment of multimodal information through fusion and interaction, the integration of heterogeneous features can still lead to varying degrees of information redundancy and interference. Leveraging the SimSiam network [33], we facilitated the refined representations derived from disentanglement to learn from the original multimodal features in reverse:

$$H_m = \mathrm{stack}_1[H_t, H_v, H_a] \quad (21)$$

Here, $i \in \{t, a, v\}$ represents the reconstruction of the common feature decoder applied to the three modalities. This strategy effectively ensures the precise extraction of key information and significantly reduces information loss, thereby strengthening the ability of the model to process multimodal data and ensuring the completeness and accuracy of feature representations. Through this holistic approach, we can effectively address issues related to missing information and redundancy while preserving the richness and diversity of multimodal data. As illustrated in Fig. 3, the SimSiam framework consists of a multilayer perceptron (MLP)-based projector p(Â·) and an MLP-based predictor q(Â·):

Projector: Used to map the input data to a latent space.

Predictor : Used to predict the latent representations.

The input representations of two different views undergo initial processing by the projector, following which the predictor maps the intermediate representation of one view to that of the other view. Additionally, we utilized a gradient-stopping operation, $sg(\cdot)$, to prevent model collapse. By leveraging the structural advantages of the SimSiam framework [34], we effectively circumvented potential issues during the learning of multimodal interaction features, ensuring the stability and effectiveness of the model learning process. These two components collectively constitute a self-supervised learning framework for acquiring useful feature representations. We conducted mutual learning on the fused and generated features using the SimSiam network as follows:

$$L_{mfr} = \frac{1}{2}[D(q(p(H^m))), sg(p(F^m)), D(sg(p(H^m))) + q(p(F^m))] \quad (22)$$

where the symmetrical SimSiam loss is defined as follows:

$$D(x, y) = \frac{-x^\top y}{\|x\|_2 \|y\|_2} \quad (23)$$

3.5. Overall loss function

For classification tasks, we employ the cross-entropy loss as loss function:

$$\mathcal{L}_{\text{task}} = -\frac{1}{n}\sum_{i=1}^{n} Y_i \cdot \log Y_m. \quad (24)$$

For regression tasks, we use the mean squared error loss as loss function:

$$\mathcal{L}_{\text{task}} = \frac{1}{n}\sum_{i=1}^{n} \|Y_i - Y_m\|_2^2 \quad (25)$$

where Y_i denotes the ground truth, and n denotes the number of samples in a batch. In conjunction with the design of our model described above, we adopted different loss components, including a common feature of learning loss L_{com}, a private feature of learning loss L_{pri}, an unimodal feature reconstruction loss L_{ufr}, and a multimodal feature reconstruction loss L_{mfr}. The combined training loss is as follows:

$$L_{\mathrm{all}} = L_{\mathrm{Task}} + \alpha \cdot L_{\mathrm{com}} + \beta \cdot L_{\mathrm{pri}} + \gamma \cdot L_{\mathrm{ufr}} + \delta \cdot L_{\mathrm{mfr}} \quad (26)$$

4. Experiments

4.1. Datasets

In this section, we outline experiments conducted on three benchmark datasets widely employed in MSA research, with a brief introduction to each dataset provided below. We enumerate the number of sentiment categories for each dataset in Table 1.

CMU-MOSI(Carnegie Mellon University Multimodal Opinion Sentiment and Emotion Intensity Dataset) [35]: This dataset comprises YouTube video clips featuring speakers delivering monologues expressing their personal opinions and feelings on various topics. The modified dataset consists of 2199 utterance-level video segments created by 89 speakers from 93 videos. Each video is annotated with emotion levels and manually assigned emotional intensity labels ranging from −3 (very negative) to 3 (very positive).

CMU-MOSEI(CMU Multimodal Opinion Sentiment and Emotion Intensity Dataset) [36]: This is an enhanced version of CMU-MOSI and offers a more extensive dataset compared to previous iterations, featuring over 65 h of video material and lectures from more than 1000 speakers. It includes 23,453 manually annotated sentence-level video paragraphs addressing 250 topics.

CH-SIMS [37]: This dataset serves as the primary benchmark dataset for Chinese unimodal and multimodal sentiment analysis, comprising 2281 sentence-level video segments extracted from 60 TV series. Unlike the preceding datasets, CH-SIMS incorporates multimodal as well as unimodal annotations, enabling researchers to explore interactions between different modalities or conduct unimodal sentiment analysis using independent unimodal annotations. However, for experiments pertinent to this study, only multimodal data annotations were utilized.

Table 1
Dataset statistics and sentiment label distributions for CMU-MOSI, CMU-MOSEI, and CH-SIMS.

Dataset	Sentiment			Data Split		
	Positive	Neutral	Negative	Train	Valid	Test
CMU-MOSI	1080	96	1023	1284	229	686
CMU-MOSEI	11 220	4986	6571	16 326	1871	4659
CH-SIMS	1238	345	698	1368	456	457

4.2. Evaluation metrics

To ensure a fair comparison, we employed multiple evaluation metrics for regression tasks conducted on the CMU-MOSI and CMU-MOSEI datasets, including mean absolute error (MAE), Pearson correlation coefficient (Coor), seven-class accuracy (acc-7), and two-class accuracy (acc-2). We used the "–/–" notation to distinguish between negative/nonnegative and negative/positive classifications, excluding zero values. For the evaluation of the CH-SIMS dataset, we selected five-class and two-class accuracies, Coor, MAE, and F1 scores as evaluation metrics. This comprehensive approach demonstrates the model's effectiveness in handling various sentiment labels.

4.3. Implementation details

Our model was constructed using the PyTorch framework, and it underwent efficient training utilizing Nvidia Tesla V100 GPUs. We employed the adaptive moment estimation (Adam) optimizer to facilitate this process. In our pursuit of the optimal configuration, we conducted a random search to fine-tune the model hyperparameters. This rigorous selection process determines that the ideal batch sizes are 32 for CMU-MOSI and CH-SIMS and 16 for CMU-MOSEI, effectively balancing processing efficiency and memory utilization. The learning rates were set at $1e-3$ for CMU-MOSI, $1e-4$ for CMU-MOSEI, and $2e-5$ for CH-SIMS. Additionally, we utilized an early stopping mechanism to prevent overfitting, ensuring that the training process halts when the model reaches its peak performance. In particular, training ceased after 8 epochs without significant improvement.

4.4. Baselines

To comprehensively evaluate the performance of our proposed method, we compared it against various popular MSA approaches and methods centred around multimodal feature decoupling. The benchmarks are as follows:

BC-LSTM [38]: This method utilizes a bidirectional long short-term memory (LSTM) network to create context-aware utterance-level representations tailored for sentiment analysis.

Tensor Fusion Network(TFN) [6]: TFN employs threefold Cartesian product-based interactions to capture both intramodality and intermodality dynamics through end-to-end fusion.

Low-Rank Multimodal Fusion(LMF) [13]: LMF incorporates low-rank factors specific to each modality, achieving exceptional efficiency.

Multimodal Transformer(MulT) [39]: MulT utilizes directed pairwise cross-modal attention to capture correlations between different modalities.

Intermodal Co-Attention Correlation Network(ICCN) [7]: ICCN learns correlations between all three modalities through deep canonical correlation analysis.

Multimodal Adaptation Gate for BERT(MAG-BERT) [14]: MAG-BERT integrates a multimodal adaptation gate (MAG) into different layers of the BERT backbone network.

Modality-Invariant and -Specific Representations for Multimodal Sentiment Analysis (MISA) [11]: This method partitions each modality into modality-invariant and modality-specific subspaces, providing a comprehensive perspective on multimodal data.

Learning Modality-Specific Representations with Self-Supervised Multitask Learning for Multimodal Sentiment Analysis (Self-MM) [18]: This approach employs unimodal supervision via an unimodal label generation module based on self-supervised learning.

Multimodal Information Modulation(MMIM) [10]: MMIM introduces a hierarchical mutual information maximization framework that enhances mutual information between input unimodal pairs and between the multimodal fusion result and the unimodal inputs.

Transformer-Based Feature Reconstruction Network(TFR-Net) [40]: This network integrates intramodal and cross-modal attention mechanisms and feature reconstruction modules to address missing features.

Feature-Disentangled Multimodal Emotion Recognition (FDMER) [12]: FDMER constructs common and private features and introduces a modality discriminator to guide the parameter learning processes of both common and private encoders adversarially.

Transformer-Based Multimodal Binding and Learning Model(TMBL) [41]: TMBL combines bimodal and trimodal binding mechanisms, fine-grained convolution modules and employs similarity and dissimilarity losses to facilitate model convergence.

Text-Enhanced Transformer Fusion Network (TETFN) [42]: TETFN employs text-oriented pairwise cross-modal mappings and text-based multihead attention, retaining diverse information among various modalities through unimodal label prediction.

Adaptive Language-guided Multimodal Transformer(ALMT) [43]: ALMT introduces an adaptive hypermodality learning (AHL) module that learns the irrelevance and conflict information contained in visual and audio features under the guidance of language features to obtain a complementary and joint multimodal representation.

4.5. Results and discussion

4.5.1. Comparison with baseline methods

The proposed FDR-MSA model marks a significant leap forward in MSA, surpassing existing methodologies. To ensure fair comparisons and reliable outcomes, we conducted rigorous tests with five independent runs for each model, maintaining uniform hyperparameter settings. Baseline results were sourced from the original publications, and we meticulously confirmed these results through replication. As shown in Table 2, our FDR-MSA model achieved an impressive score of 0.703 on the CMU-MOSI dataset. Additionally, it excelled in various classification metrics, including accuracy and the F1 score. On the CMU-MOSEI dataset (Table 3), FDR-MSA again demonstrated its superiority, achieving an MAE of 0.521, surpassing the MAEs of all other compared baselines. The model's strength lies in its innovative approach to handling private and common features, effectively reducing information redundancy and loss. This efficacy extends to the CH-SIMS benchmark, as depicted in Table 4, where FDR-MSA consistently outperformed other baselines. For instance, it achieved an MAE reduction of 0.028 compared to the second-best model, Self-MM, along with a 2.6% accuracy improvement. In summary, these results underscore the efficacy of FDR-MSA, representing a significant advancement in the intricate field of MSA and paving the way for future developments in this domain.

4.6. Quantitative analysis

4.6.1. Impact of key components

Table 5 presents the detailed ablation experimental results obtained for each component within the proposed FDR-MSA framework.

(1) Feature Disentanglement Learning

A comparative analysis of the performance achieved with different components notably indicates a significant decrease in model efficacy when removing the private feature learning and common feature learning components (denoted by "$w/o\ L_{private}$" and "$w/o\ L_{common}$",

Table 2
Performance comparison on the CMU-MOSI benchmark. All other results were reproduced using publicly available source codes and original hyper-parameters under the same settings.

Model	MAE (↓)	Acc2 (↑)	Acc7 (↑)	F1 (↑)	Corr (↑)
BC-LSTM[c]	1.709	73.9/–	28.7	73.9/–	0.581
TFN[b]	0.901	–/80.8	34.9	–/80.7	0.698
LMF[b]	0.917	–/82.5	33.2	–/82.4	0.695
MulT[a]	0.861	81.5/84.1	–	80.6/83.9	0.711
ICCN[c]	0.860	–/83.0	39.0	–/83.0	0.710
MAG[a]	0.712	84.2/86.1	–	84.1/86.1	0.796
Self_MM[a]	0.712	82.5/84.8	45.8	82.8/84.6	0.717
MMIM	**0.700**	84.1/86.1	46.7	84.0/86.0	0.712
AMML	0.723	–/84.9	46.3	–/84.8	0.792
TFR-Net	0.754	–/84.1	–	–	0.721
MISA[c]	0.783	81.8/83.4	42.3	81.7/83.6	0.761
FMDER[b]	0.724	84.6/–	44.1	84.7/–	0.788
TETFN	0.717	84.05/86.10	–	83.83/86.07	**0.800**
TMBL	0.867	81.78/83.84	36.3	82.41/84.29	0.762
ALMT	0.72	82.94/84.91	**47.08**	83.07/84.97	0.783
FDR-MSA(ours)	0.703	**85.57/86.28**	46.84	**85.73/86.31**	0.796

[a] Denotes results from [18].
[b] Denotes results from [12].
[c] Denotes results from [11].

Table 3
Performance comparison on the CMU-MOSEI benchmark. All other results were reproduced using publicly available source codes and original hyper-parameters under the same settings.

Model	MAE (↓)	Acc2 (↑)	Acc7 (↑)	F1 (↑)	Corr (↑)
TFN[b]	0.593	–/82.5	50.2	–/82.1	0.700
LMF[b]	0.623	–/82.0	48.0	–/82.1	0.677
MulT[a]	0.580	–/82.5	–	–/82.3	0.703
ICCN[c]	0.565	–/84.2	51.6	–/84.2	0.713
Self_MM[a]	0.529	82.7/85.0	53.5	83.0/84.9	0.767
MMIM	0.526	82.2/86.0	54.2	82.7/86.0	0.772
AMML	0.614	–/85.3	52.4	–/85.2	0.776
MISA[c]	0.555	83.6/85.5	52.2	83.8/85.3	0.765
FMDER[b]	0.536	–/86.1	54.1	**85.8**	0.773
TETFN	0.551	84.25/85.18	–	84.18/85.27	0.748
TMBL	0.545	84.23/85.84	52.4	84.87/85.92	0.766
ALMT	0.533	82.23/85.66	52.82	81.89/85.79	0.777
FDR-MSA(ours)	**0.521**	**84.76/86.32**	**56.7**	84.7/85.6	**0.791**

[a] Denotes results from [18].
[b] Denotes results from [12].
[c] Denotes results from [11].

Table 4
Performance comparison on the CH-SIMS benchmark. All other results were reproduced using publicly available source codes and original hyper-parameters under the same setting.

Model	MAE (↓)	Acc-2 (↑)	Acc-5 (↑)	F1 (↑)	Corr (↑)
TFN[b]	0.437	77.1	–	–	0.582
LMF[b]	0.483	77.4	–	77.4	0.578
MulT[a]	0.442	78.2	40.0	78.5	0.581
Self_MM[a]	0.411	78.6	43.1	78.6	0.601
MMIM	0.422	78.3	42.0	78.2	0.597
ALMT	0.414	78.99	40.9	79.0	0.602
FDR-MSA(ours)	**0.383**	**81.2**	**43.7**	**80.3**	**0.627**

[a] Denotes results from [18].
[b] Denotes results from [12].

respectively). This outcome underscores the importance of feature disentanglement in achieving improved prediction accuracy and validates the original intent of our model design, namely, to prevent information loss by effectively capturing relevant features. When the "private" and "common" configuration rules are removed, the learning process of the model degrades in terms of finely decoupling heterogeneous data to extract desired attributes, resulting in declines across all performance metrics. The "common" component plays a pivotal role in enabling

Table 5
Comprehensive comparison on the CMU-MOSI, CMU-MOSEI, and CH-SIMS benchmarks. The symbol "(–)" indicates the omission of a component.

Component	CMU-MOSI				
	MAE (↓)	Corr (↑)	Acc-7 (↑)	Acc-2 (↑)	F1 (↑)
$w/o\ L_{private}$	0.732	0.767	44.01	83.46	83.28
$w/o\ L_{common}$	0.753	0.745	43.75	82.89	82.97
$w/o\ L_{ufr}$	0.750	0.786	44.31	83.06	83.11
$w/o\ L_{mfr}$	0.738	0.787	44.57	83.59	83.26
All	0.703	0.796	46.84	85.57	85.73
Component	**CMU-MOSEI**				
	MAE (↓)	Corr (↑)	Acc-7 (↑)	Acc-2 (↑)	F1 (↑)
$w/o\ L_{private}$	0.574	0.726	53.35	82.99	82.68
$w/o\ L_{common}$	0.565	0.715	52.06	82.16	81.95
$w/o\ L_{ufr}$	0.573	0.723	52.3	81.91	81.73
$w/o\ L_{mfr}$	0.536	0.764	53.9	82.86	82.45
All	0.521	0.791	56.7	84.76	84.7
Component	**CH-SIMS**				
	MAE (↓)	Corr (↑)	Acc-5 (↑)	Acc-2 (↑)	F1 (↑)
$w/o\ L_{private}$	0.412	0.579	42.06	79.09	78.87
$w/o\ L_{common}$	0.465	0.565	41.38	78.37	78.21
$w/o\ L_{ufr}$	0.417	0.591	41.7	78.2	77.9
$w/o\ L_{mfr}$	0.459	0.573	41.6	77.9	77.4
All	0.383	0.627	43.7	81.2	80.3

the model to learn data commonalities across different modalities, which are essential for capturing generic emotional traits. Without the "private" component, the model struggles to accurately comprehend private emotional information and nuances within modalities, adversely affecting the precision of its sentiment classification results.

(2) Feature Reconstruction Learning

While the absence of L_{ufr} and L_{mfr} does not induce as severe detrimental effects as the lack of $L_{private}$ and L_{common}, that both UFR and MFR play significant roles in enhancing the key performance indicators of the constructed model. Upon removing these two modules from the model, we noted slight decreases in evaluation metrics such as Acc2, F1, and Corr. Our dual reconstruction process ensures that no valuable or critical information is lost during the decoupling and fusion processes. Furthermore, the performance degradation caused by the removal of L_{mfr} is generally less pronounced than the losses caused by the removal of L_{ufr} on the CMU-MOSI and CMU-MOSEI datasets; however, on the CH-SIMS dataset, the absence of L_{mfr} led to a more noticeable decline in performance. The experimental data showed that both components were indispensable in sentiment analysis tasks, although their importance levels may vary across different datasets and contexts. The joint utilization of these components can greatly favour the model.

Our framework accurately captures and analyses complex emotional features, making a substantial contribution to advancing MSA research. These findings systematically validate the distinct contributions of each component, illustrating that the exclusion of any element results in performance degradation, which is consistently observed across all evaluation metrics and datasets. This pattern unequivocally establishes that the integration of feature disentanglement and reconstruction significantly boosts the overall performance of our model.

4.6.2. Importance of the various modalities

After analysing the performance of multimodal combinations as presented in Table 6, we observed that the model performs optimally under the "all" setting, which incorporates text (T), visual (V), and auditory (A) modalities, demonstrating the lowest MAEs and the highest accuracy and F1 scores. The inclusion of additional modality data enhances the model's performance, affirming the scientific robustness and reliability of our approach and validating the effectiveness of our multimodal representation learning strategy in comprehensively

Table 6
Comparison of the CMU-MOSI, CMU-MOSEI, CH-SIMS, and experimental results for different modal settings. V, A, and T indicate visual, acoustic, and textual modalities.

Modalities	CMU-MOSI				
	MAE (↓)	Coor (↑)	Acc-7 (↑)	Acc-2 (↑)	F1 (↑)
A&V	0.791	0.705	43.45	80.75	79.67
T&V	0.783	0.732	44.08	82.15	82.02
T&A	0.75	0.749	44.9	82.31	82.13
All	0.703	0.796	46.84	85.57	85.73
Modalities	CMU-MOSEI				
	MAE (↓)	Coor (↑)	acc-7 (↑)	acc-2 (↑)	F1 (↑)
A&V	0.61	0.614	43.8	78.49	78.04
T&V	0.543	0.709	46.9	82.92	81.82
T&A	0.574	0.693	46.3	81.65	80.53
ALL	0.521	0.791	56.7	84.76	84.7
Modalities	CH-SIMS				
	MAE (↓)	Coor (↑)	Acc-5 (↑)	Acc-2 (↑)	F1 (↑)
A&V	0.498	0.54	40.2	76.5	76.3
T&V	0.454	0.583	41.7	78.2	77.9
T&A	0.436	0.591	42.6	78.3	78.1
All	0.383	0.627	43.7	81.2	80.3

Table 7
Comparison of the experimental outcomes for various disentangle strategies applied to the FDR-MSA model.

	CMU-MOSI		CMU-MOSEI	
	Acc-2	F1	Acc-2	F1
−MISA	83.32	82.51	83.78	83.92
−FDMER	84.83	85.01	86.37	86.45
Ours	85.57	85.73	84.76	84.7

understanding emotions and accurately predicting them. When comparing the performances of different modal combinations, we found that the textual modality holds the most significance among the three modalities. In scenarios lacking textual information, modal combinations comprising only visual (V) and auditory (A) features generally exhibit lower performance across the testing datasets compared to the full modality combination. This indicates that while other modalities can convey emotional signals, the absence of explicit semantic information from text adversely impacts the model's emotion prediction performance.

4.6.3. Different disentanglement strategies

To validate the effectiveness of our FDR-MSA decoupling strategy, we substituted the decoupling component within the model with other decoupling algorithms. Compared to other representative and advanced methods for extracting features from different modalities, as documented in current literature [11,12], our decoupling strategy demonstrated notable competitiveness. This not only confirms the rationale and superiority of our proposed method but also empirically validates our initial intention to preserve crucial information throughout the decoupling process (see Table 7).

4.7. Visualization analysis

4.7.1. Analysis of loss trends

Loss metrics are pivotal in our research, quantifying and evaluating the effectiveness of the developed model in representation learning and reconstruction tasks. The loss index set $L_{com}, L_{pri}, L_{uft}, L_{mfr}$ is crucial for measuring the model's performance in multimodal representation learning and feature reconstruction tasks, as depicted in Fig. 4. Monitoring the evolution trends of these loss values throughout the training process provides insights into the model's performance consistency with our theoretical design. As shown in the charts of Fig. 4, each designed loss indicator exhibits a significant downward trend as the training duration extends. This trend reflects the model's improvement in performance during the optimization process, demonstrating its increasing ability to capture and comprehend representations across different modalities. Particularly in the early training stages, some loss curves may show fluctuations, indicating the model's adaptation to various data characteristics. However, as training progresses, these fluctuations diminish, and the loss curves become smoother, signifying the gradual improvement in the stability and performance of the model. This observed trend is crucial for validating the model design, confirming that the model follows the learning process as per the preset framework and objectives.

4.7.2. Visualization of common and private representations

Gaining insight into the subspaces of different representations is paramount, particularly in tasks involving multimodal representational learning. Models tasked with handling intricate data from diverse modalities such as text, audio, and visual inputs need to demonstrate and interpret the representations acquired during training to understand their functionality fully. As illustrated in Fig. 5, we scrutinize the model's capability for representation learning across the three primary datasets utilized in our study. Here, (C_t, C_v, C_a) encapsulate the common features learned by the model, while (P_t, P_v, P_a) represent private features, respectively.

The visualized outcomes illustrate substantial aggregation and overlap among our common features, underscoring our achievement in capturing shared characteristics across modalities and ensuring the consistency of these common features. Conversely, the distinct separation of private features suggests our model's proficiency in acquiring modality-specific and distinctive information. Typically, overlaps in visualized representations indicate redundancy or strong correlations within the learned information. However, our results, characterized by evident gaps, signify clear disparities and diversity in the information captured for each modality, thereby showcasing the robust discriminative capabilities of the model. Adjusting the weights of the learning rules for common and private features induces significant changes in the feature distribution learned by the model. Common features exhibit some degree of overlap, while private features tend toward separation, indicating scope for improvement toward an ideal state characterized by clear delineation and tight clustering.

4.7.3. Analysis of the impact of attention

Boxplots were used to analyse the attention distributions of common and private features in the test sets. The results presented in Fig. 6 demonstrate the consistent allocation of attention by the model across all test sets. This indicates that our model is robust in terms of adjusting the contributions of different features and can adaptively tune itself according to the characteristics of various datasets. The attention weights yielded by the common features consistently exceeded those of the private modality features, with a significantly higher median. This trend highlights the central role of common features in ensuring consistency during the sentiment analysis process. It confirms the efficiency of our model with respect to the optimization of common features. Across different datasets, the attention weights produced by the private features vary. Enhancing common features through an attention mechanism facilitates the acquisition of balanced sentiment prediction outcomes. These visual analyses validated the dominant position of common features in MSA and revealed the subtle yet crucial role of private features in enhancing sentiment recognition. The adaptive attention mechanism of the proposed model demonstrated its ability to identify and utilize key features to produce accurate sentiment prediction results. The consistent results observed across various datasets further emphasize the robustness and adaptability of the model.

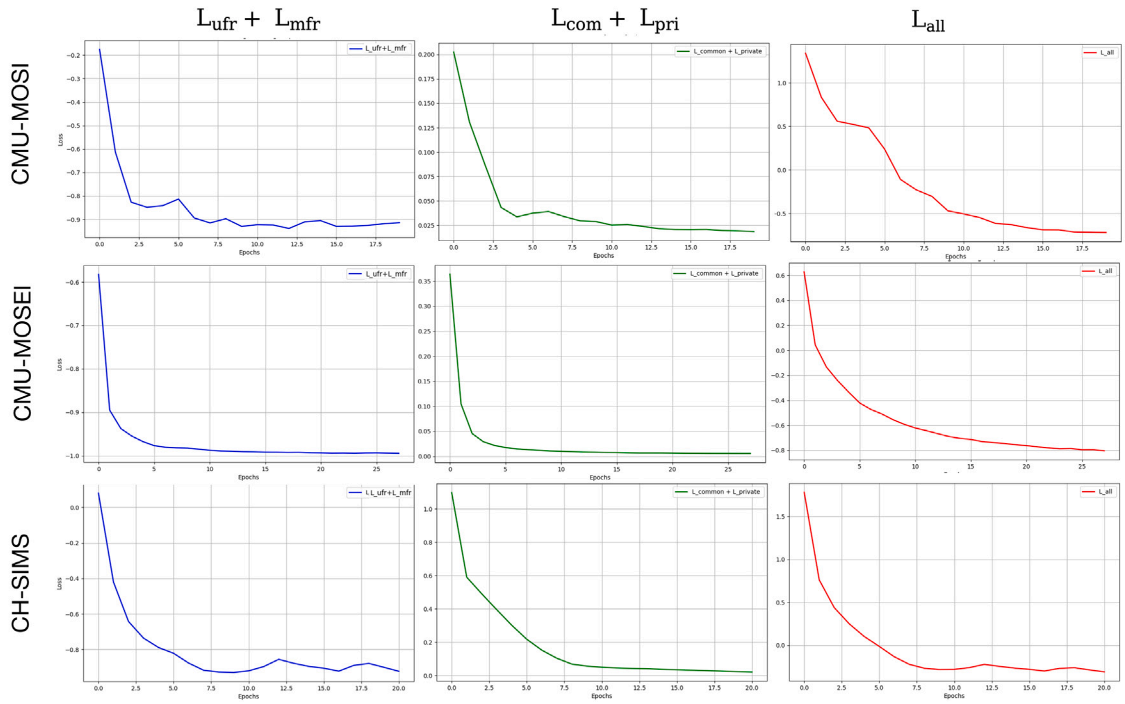

Fig. 4. Visualization of unimodal and multimodal feature reconstruction $L_{\text{ufr}} + L_{\text{mfr}}$, common and private features $L_{\text{com}} + L_{\text{pri}}$, and overall loss L_{all} tracking throughout the training process.

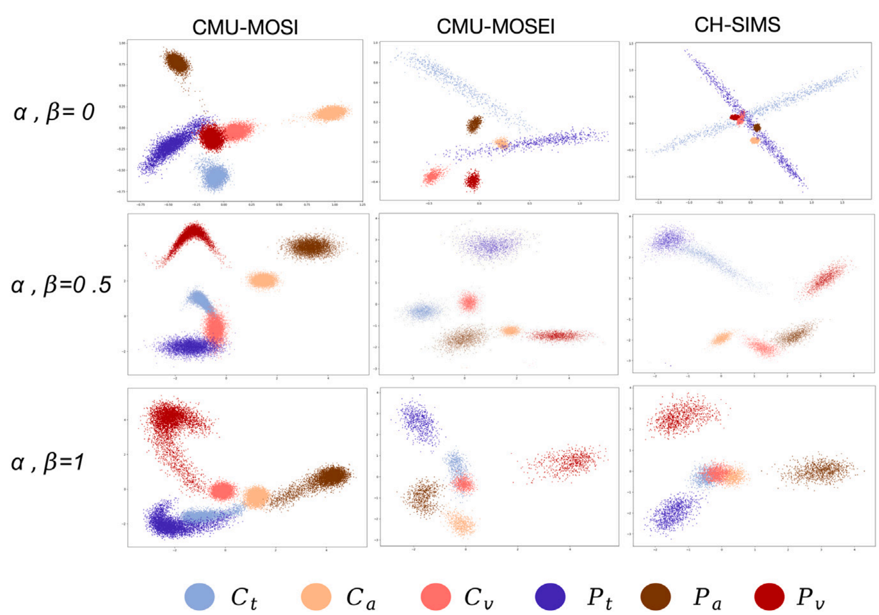

Fig. 5. Visualization of test set features across three benchmarks, with α and β indicating weights of common and private losses respectively.

4.7.4. Case study

This study showcases the effectiveness of multimodal sentiment analysis models in real-world scenarios, as depicted in Fig. 7. We meticulously selected representative examples from the CMU-MOSEI and CH-SIMS datasets for detailed analysis. In the CMU-MOSEI dataset, the FDR-MSA model demonstrated remarkable performance, yielding a prediction value of 0.03, closely aligning with the actual sentiment label of 0. This accuracy surpassed the highest prediction values achieved by the MISA and Self-MM models, which were 0.63 and 0.14, respectively. When applied to the Chinese dataset CH-SIMS, while all models could discern negative emotions, the FDR-MSA model's prediction value of −0.45 closely matched the actual label of −0.8. This indicates superior precision in emotion recognition compared to other baseline models. We also conducted specific evaluations of the model's performance without shared and private feature decoupling ($L_{pri\&com}$) and feature reconstruction ($L_{ufr\&mfr}$). The results revealed a significant enhancement in the model's ability to capture emotions with these techniques in place, further underscoring the effectiveness and necessity of these methods in handling complex multimodal emotional data.

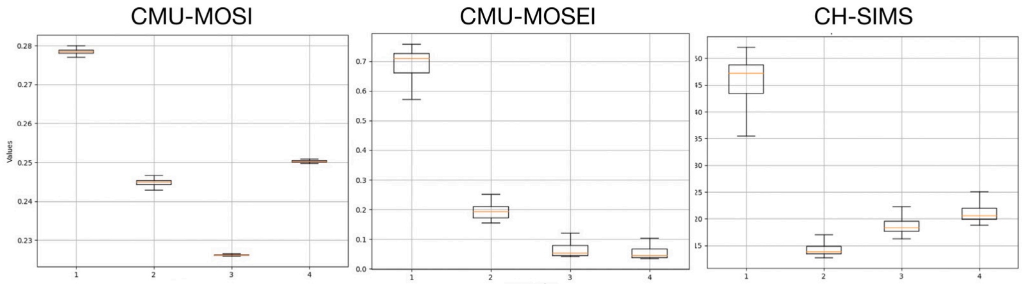

Fig. 6. Illustration of the distributions of common and private features derived from the adaptive attention fusion module on the test sets of the three benchmarks. The features are arranged in ascending order of their feature indices: common features, private text features, private audio features, and private video features.

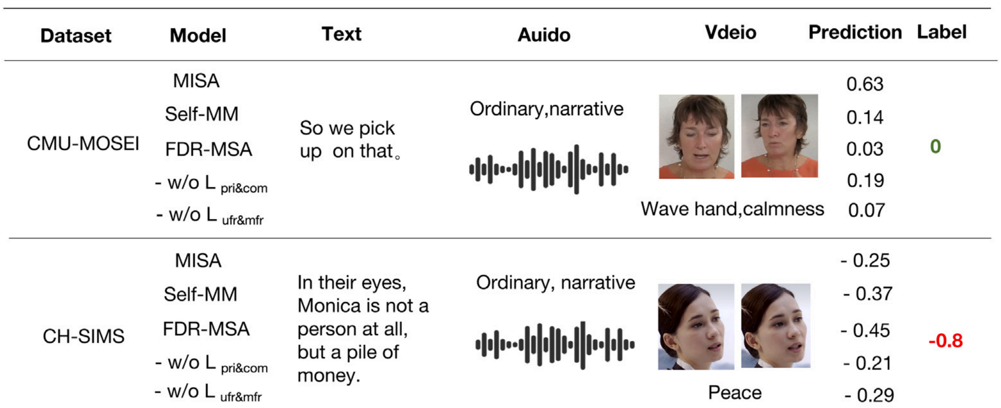

Fig. 7. Comparison of instances from the CMU-MOSEI and CH-SIMS datasets showing performance of baseline models Self-MM, MISA, and our FDR-MSA. The table details performance impacts post-ablation: 'w/o L pri&com' denotes decoupling function removal, and 'w/o L ufr&mfr' denotes reconstruction function removal.

5. Conclusion

In this study, we introduce an innovative framework for Multimodal Sentiment Analysis (MSA) called FDR-MSA, built upon feature decoupling and reconstruction techniques. This framework aims to optimize the representation learning process essential for MSA tasks. Central to this framework is the optimization of multimodal data representation learning through decoupling, integration, and reconstruction strategies. We meticulously decouple heterogeneous multimodal features and employ a learning strategy that capitalizes on their shared and distinct characteristics. By reconstructing both multimodal and unimodal features, we mitigate potential information loss or distortion during the decoupling process, ensuring the consistency and diversity of essential information. This approach facilitates effective interaction and integration of these features within a latent space, compensating for potential information loss and uncovering subtle differences and interactions among different modalities. Our model offers a fresh perspective for implementing multimodal representation learning. Extensive experiments across three datasets consistently demonstrate that our model achieves state-of-the-art performance. Visual analysis of the experiments further validates the alignment of the model's performance with our experimental design expectations. Leveraging multimodal representations, our model reliably completes sentiment analysis tasks. In future work, we intend to explore the impact of multimodal reconstruction on representation learning under noisy and missing data conditions. Additionally, we aim to investigate the effects of modal decoupling and reconstruction on task performance. These research pursuits will advance the field of multimodal learning, enhancing the adaptability and accuracy of our model for use in complex environments.

CRediT authorship contribution statement

Yao Fu: Writing – original draft, Visualization, Methodology, Conceptualization. **Biao Huang:** Writing – review & editing, Formal analysis, Conceptualization. **Yujun Wen:** Funding acquisition. **Pengzhou Zhang:** Supervision, Resources.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: YujunWen reports financial support was provided by National Key R&D Program of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Comparison with LLMs

Large language models (LLMs) have showcased remarkable capabilities in processing intricate language expressions, making them valuable assets for exploring human sentiment comprehension. GPT-4V stands at the forefront of multimodal LLMs in terms of its capabilities, rendering it an ideal candidate for comparative assessments to gauge LLM

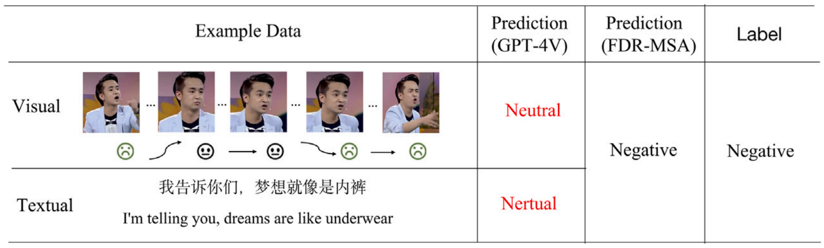

Fig. A.8. Examples of incorrect predictions by GPT-4V and explanations of its sentiment evaluation process for these cases.

Table A.8
Performance comparison with GPT-4V.

Model	CMU-MOSI	CH-SIMS
FDR-MSA (T&V&A)	**86.28**	79.7
FDR-MSA (T&V)	83.01	71.85
GPT-4 (Text only)	82.32[a]	70.07[a]
GPT-4 (T&V)	80.43[a]	**81.24**[a]

[a] Results from [44].

performance in MSA tasks. However, GPT-4V lacks native support for video inputs, necessitating the extraction of frames from videos at fixed intervals, which are subsequently fed into GPT-4V sequentially. Additionally, GPT-4V does not incorporate audio modalities into its multimodal task setups. Thus, we compare the performances of GPT-4V, GPT-4V (text only), and FDR-MSA across two and three modalities, as summarized in the Table A.8.

During the case study, shown in Fig. A.8, we noted that while GPT-4V benefits from its extensive training corpus, its research focus primarily centres on the general domain. Our evaluation of GPT-4V indicates that when tackling multimodal emotion perception tasks, the model does not adopt a multimodal fusion approach to integrate multimodal sentiment features. Instead, it tends to process emotions independently within each modality, resembling traditional decision fusion methods. This approach differs from the natural human strategy of intelligently integrating information from multiple modalities to perceive emotions. In contrast, research methods developed for multimodal emotion analysis closely emulate human emotion perception mechanisms and can effectively address the limitations of models such as GPT-4V in complex scenarios. This highlights the importance of utilizing supervised learning methods to more effectively integrate emotion information.

Furthermore, despite its extensive training in the general domain, GPT-4V may encounter challenges in detecting subtle emotional changes. For example, in the text “Dreams are like underwear”, GPT-4V struggles to accurately discern its negative emotion, indicating potential shortcomings in handling emotional expressions within specific cultural contexts. In contrast, supervised models specifically tailored for sentiment classification benefit from specialized training processes and demonstrate effective identification results.

Appendix B. Comparison with encoders

In current research on MSA, the BERT model is widely used for extracting textual features in multimodal emotion analysis. To explore whether more advanced models can provide more robust semantic representation capabilities, we conducted experiments using RoBERTa [45] instead of BERT. RoBERTa has demonstrated exceptional performance across various natural language processing tasks. The experiments compared sentiment analysis models using BERT and RoBERTa encoders. The results indicated that RoBERTa slightly enhanced the models’ ability to handle complex emotional information due to its superior semantic understanding capabilities, underscoring the importance of adopting advanced text encoding techniques to enhance model performance (see Table B.9).

Table B.9
Performance comparison across different datasets.

	CMU-MOSI		CMU-MOSEI		CH-SIMS	
	Acc-2	F1	Acc-2	F1	Acc-2	F1
FDR-MSA	85.57	85.73	84.76	84.7	81.2	80.3
+Roberta	85.69	85.92	85.01	85.12	81.4	80.35

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